Report: Online Retail Customer Value Analysis and Repurchase Behavior Prediction

Course name: INFO442 Data Science Projects

Student: Zhang Wubin

Zhang Jiaming

Wang Yunhe

Chen Xuanzhi

An Zelin

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Catalogue

1. Introduction

Customer retention has become a key competitive strategy in modern e-commerce. As online retail markets become increasingly saturated, acquiring new customers requires substantial marketing investment, while retaining existing customers often generates significantly higher returns. Therefore, predicting whether a customer is likely to make another purchase provides valuable information for personalized marketing, inventory planning, and customer relationship management.

This project uses the Online Retail dataset collected from a UK-based online gift retailer between December 2010 and December 2011. The dataset contains detailed transaction records, including customer IDs, purchased products, transaction dates, quantities, prices, and countries. By analyzing customer purchasing behavior and applying machine learning techniques, this project aims to identify customers who are most likely to make another purchase within the next thirty days.

Unlike purely theoretical machine learning exercises, this project follows the complete data science lifecycle. Starting from exploratory data analysis, business insights are extracted from historical transaction data. These insights guide feature engineering and predictive model development. Finally, a practical deployment strategy is proposed to demonstrate how predictive analytics can be integrated into business operations.

2. Business Problem and Project Objectives

2.1 Business Problem

The primary business challenge addressed in this project is customer repurchase prediction.

Many customers only make one purchase and never return, while loyal customers repeatedly contribute revenue over time. However, marketing budgets are limited, making it inefficient to target every customer equally. Businesses therefore require predictive models capable of identifying customers with high repurchase probability.

The prediction problem is formulated as a binary classification task:

Target Variable

Whether a customer will make at least one purchase within 30 days after the observation period.

2.2 Project Objectives

The objectives of this project are:

- Understand customer purchasing behavior through exploratory data analysis.
- Identify important business patterns affecting customer retention.
- Construct predictive customer-level features using transaction history.
- Compare different machine learning algorithms.
- Evaluate prediction performance using multiple metrics.
- Deploy the best-performing model into a practical business workflow.
- Provide marketing recommendations based on prediction results.

Achieving these objectives enables businesses to improve marketing efficiency, customer retention, and long-term profitability.

3. Dataset Description

3.1 Dataset Overview

This project uses the Online Retail dataset, which records transactions made by a UK-based online retailer specializing in gift products. The dataset covers approximately thirteen months of business operations, from December 2010 to December 2011.

After removing cancelled transactions and invalid records, the cleaned dataset contains:

- 388,804 valid transaction records
- 4,303 unique customers
- 3,663 unique products
- Total revenue exceeding £7.19 million

- Each transaction includes:
- Invoice Number
- Customer ID
- Product Description
- Quantity
- Unit Price
- Invoice Date
- Country

These variables provide comprehensive information for analyzing purchasing behavior.

3.2 Data Cleaning

Several preprocessing procedures were performed before analysis.

First, all return transactions were removed because they do not represent successful purchases.

Second, records with non-positive quantities or unit prices were discarded to eliminate abnormal transactions.

Third, missing customer identifiers were excluded because customer-level prediction requires unique customer histories.

Finally, invoice dates were converted into datetime format to facilitate temporal feature extraction.

These preprocessing steps produced a reliable dataset suitable for exploratory analysis and predictive modeling.

3.3 Business Understanding

Although the retailer serves customers from multiple countries, exploratory analysis reveals that the UK contributes the majority of total revenue. Furthermore, customer purchasing activity exhibits strong seasonal patterns, especially during the Christmas shopping period.

These characteristics suggest that customer historical behavior contains valuable information for predicting future purchases, providing the foundation for machine learning modeling in later stages of the project.

4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand customer purchasing behavior, identify business patterns, and provide insights for feature engineering and predictive modeling. The analysis focused on temporal trends, geographical distribution, product performance, and customer purchasing characteristics.

4.1 Monthly Sales Trend

The monthly transaction analysis reveals clear seasonality throughout the thirteen-month observation period. Transaction volume remained relatively stable during the first half of 2011, followed by steady growth from July onward. The highest sales volume occurred in November 2011, immediately before the Christmas holiday season.

This seasonal pattern reflects the purchasing behavior of customers preparing for holiday shopping. After the November peak, transaction volume declined sharply in December as most seasonal purchases had already been completed.

The observed seasonality suggests that customer purchasing behavior is strongly influenced by calendar events, indicating that temporal features should be considered during predictive modeling.

4.2 Daily and Weekly Purchasing Patterns

Customer purchasing activity also exhibits consistent daily and weekly patterns.

Analysis of daily transactions showed that sales fluctuate moderately during normal business operations, with no prolonged periods of inactivity. Weekly analysis indicates that more than 84% of all transactions occur on weekdays, while weekend sales are considerably lower.

This finding suggests that many customers are business purchasers or office-related buyers who tend to place orders during working days rather than weekends.

Furthermore, hourly transaction analysis demonstrates that purchasing activity is concentrated during standard business hours. The highest number of transactions occurs around noon, while almost no purchases are recorded during late-night hours. These behavioral patterns provide useful information for scheduling promotional campaigns and timing marketing communications.

4.3 Geographic Revenue Distribution

Revenue analysis by country demonstrates an extremely uneven market structure.

The United Kingdom contributes approximately 85% of total platform revenue, making it the dominant market throughout the observation period. Other European countries, including the Netherlands, Germany, France, and Ireland, contribute relatively small proportions of total revenue.

Although international markets generate less revenue, they still represent potential opportunities for future business expansion.

The concentration of revenue within a single country suggests that geographic location is an important variable affecting customer purchasing behavior and should therefore be incorporated into predictive modeling.

4.4 Product Sales Analysis

Product-level analysis identifies several products that consistently achieve high sales volumes.

The best-selling products mainly consist of inexpensive household decorations, gift items, and storage products rather than premium or luxury products. These affordable items dominate both transaction quantity and overall revenue.

This finding contradicts the common assumption that expensive products generate most business value. Instead, frequent purchases of low-cost products collectively contribute a substantial proportion of company revenue.

These purchasing preferences also indicate that product categories may contain predictive information regarding customer repurchase behavior.

4.5 Customer Purchase Behaviour

Customer purchasing frequency demonstrates considerable variation across the customer base.

Most customers make multiple purchases throughout the observation period, while only a small proportion purchase once and never return. The distribution is highly right-skewed, with a limited number of highly active customers placing exceptionally large numbers of orders.

This imbalance suggests that customer loyalty varies significantly across individuals.

The analysis further supports the application of customer-level behavioral features such as Recency, Frequency, and Monetary (RFM), which summarize historical purchasing patterns into measurable indicators.

5. Business Insights from EDA

The exploratory analysis provides several important business insights that directly motivate the predictive modeling stage.

First, customer purchasing behavior exhibits strong seasonality, particularly during the Christmas shopping period. Therefore, temporal information should be considered when interpreting model predictions.

Second, customer spending is highly concentrated among a relatively small number of loyal customers. Historical purchasing behavior appears to be an effective indicator of future purchasing intentions.

Third, geographic differences indicate that customer location influences purchasing patterns, although the UK dominates overall sales.

Finally, the observed purchasing frequency distribution suggests that repeat customers possess distinguishable behavioral characteristics that can be captured using engineered features.

Based on these findings, the project proceeds to construct customer-level features for supervised machine learning.

6. Feature Engineering

Feature engineering was performed to transform raw transaction records into meaningful customer-level variables suitable for machine learning. By aggregating transactional information into behavioural and statistical features, the model can better capture customer purchasing patterns and improve prediction performance. The constructed features include traditional RFM variables, additional statistical indicators, product preference features, and the prediction target.

6.1 Data Cleaning

Before constructing customer-level features, several data preprocessing procedures were conducted to ensure the quality and reliability of the dataset. Return transactions were removed because they do not represent actual purchasing behaviour. Records containing invalid purchase quantities or unit prices were excluded to eliminate obvious data entry errors. Transactions with missing customer identifiers were discarded since customer-level aggregation could not be performed without a valid customer ID. In addition, duplicate transaction records were identified and removed where appropriate. After these preprocessing steps, the remaining data provided an accurate representation of customers' purchase histories for subsequent feature engineering.

6.2 Customer Behaviour Features

Customer purchasing behaviour was primarily summarised using the widely adopted RFM (Recency, Frequency, Monetary) framework, which has been extensively applied in customer relationship management and purchase prediction tasks. Recency measures the number of days since a customer's most recent purchase, reflecting how recently the customer interacted with the business. Customers with lower recency values are generally considered more likely to make future purchases. Frequency represents the total number of completed purchases made by each customer during the observation period and serves as an indicator of customer loyalty and engagement. Monetary measures the cumulative amount spent by each customer and reflects the overall economic value of the customer. Customers with higher monetary values generally contribute more revenue and often exhibit stronger repurchase potential.

6.3 Additional Statistical Features

To provide a more comprehensive representation of customer purchasing behaviour, several additional statistical features were generated based on historical transaction records. These features include the average, maximum, and minimum order values, the total purchased quantity, the average quantity per transaction, the standard deviation of purchased quantities, and the number of unique products purchased by each customer. These variables describe different aspects of purchasing behaviour, including spending consistency, purchasing scale, and product diversity.

Furthermore, several derived features were constructed to capture customer behaviour from different perspectives. For example, the average monetary value per purchase was calculated by dividing the total monetary value by the purchase frequency, reflecting customers' average spending per transaction. Similarly, the average purchased quantity per transaction was included to describe typical purchasing volume. Compared with the traditional RFM variables alone, these additional statistical features provide richer information about customer purchasing patterns and are expected to improve the predictive capability of the machine learning models.

6.4 Product Preference Features

In addition to behavioural statistics, customer product preferences were incorporated into the feature set using Natural Language Processing (NLP) techniques. Product descriptions were first transformed into TF-IDF vectors to represent textual information numerically. K-Means clustering was then applied to group products with similar descriptions into thirteen product categories. For each customer, the product category with the highest purchase frequency was identified as the customer's dominant purchasing preference and used as an additional feature. This approach enables the model to capture customers' latent interests and purchasing preferences beyond simple transaction frequency or spending behaviour.

6.5 Target Variable Construction

The prediction target was defined based on customers' purchasing behaviour after the observation period. Specifically, a customer was labelled as positive if they made at least one additional purchase within thirty days following the end of the observation window. Otherwise, the customer was assigned a negative label. This definition transforms the customer retention problem into a binary supervised classification task, where a label of 1 indicates that the customer returned within thirty days and a label of 0 indicates that no subsequent purchase occurred during the same period. The resulting customer-level dataset was then used for training and evaluating the machine learning models.

7. Model Development

Following exploratory data analysis and feature engineering, supervised machine learning models were developed to predict whether a customer would make another purchase within the next thirty days. Since the prediction target is binary, this task was formulated as a binary classification problem.

Two widely adopted machine learning algorithms were selected for comparison: Logistic Regression and Random Forest. These models were chosen because they provide a balance between interpretability and predictive capability, making them suitable for customer behaviour prediction.

7.1 Model Selection

Two supervised machine learning algorithms were selected to predict customer repurchase behaviour: Logistic Regression and Random Forest. These models were chosen because they represent two widely used classification approaches with different modelling capabilities, allowing their predictive performance and interpretability to be compared.

Logistic Regression was selected as the baseline model due to its simplicity, computational efficiency, and strong interpretability. Unlike more complex algorithms, Logistic Regression estimates the probability that a customer belongs to the repurchase class by modelling the relationship between explanatory variables and the log-odds of repurchase. The estimated coefficients provide valuable business insights by indicating whether each feature has a positive or negative influence on customer retention. To reduce the impact of class imbalance, the model was trained using the class_weight='balanced' parameter. Logistic Regression is particularly suitable for structured tabular data because it trains quickly, produces stable predictions, and allows business stakeholders to understand how individual variables influence customer repurchase behaviour.

Random Forest was selected as the second model because it can capture complex nonlinear relationships that cannot be modelled effectively by linear algorithms. As an ensemble learning method, Random Forest constructs multiple decision trees using bootstrap sampling and combines their predictions through majority voting. This ensemble strategy generally improves prediction accuracy while reducing the risk of overfitting. In this project, the model was configured with approximately 200 decision trees, while the maximum tree depth was controlled to limit model complexity. Similar to Logistic Regression, balanced class weights were applied during training to alleviate prediction bias caused by class imbalance. In addition to its ability to model nonlinear customer behaviour, Random Forest automatically captures interactions among variables, is robust to noisy features, and provides feature importance rankings that help interpret model predictions.

7.2 Model Training

After customer-level features had been constructed, the dataset was randomly divided into training and validation sets using an 80:20 split. The training dataset was used to fit both machine learning models, while the validation dataset was reserved exclusively for performance evaluation. Although a chronological train-test split would more closely resemble real-world forecasting scenarios, the available dataset covered only one year of transaction records. Consequently, a random split was adopted for demonstration purposes while acknowledging its limitations.

The overall modelling workflow consisted of several sequential steps. First, customer-level behavioural and statistical features were generated from the cleaned transaction data. Next, the binary target variable indicating whether a customer repurchased within thirty days was constructed. The resulting dataset was then divided into training and validation subsets. Logistic Regression and Random Forest were subsequently trained using the same training data, and their predictive performance was evaluated using multiple classification metrics on the validation dataset.

7.3 Evaluation Metrics

Several complementary evaluation metrics were adopted to assess model performance from different perspectives. Because customer repurchase prediction is an imbalanced binary classification problem, relying solely on a single metric such as accuracy may provide an incomplete assessment of model quality.

The Area Under the Receiver Operating Characteristic Curve (ROC-AUC) was used to evaluate the model's ability to distinguish customers who would repurchase from those who would not across different classification thresholds. A higher ROC-AUC value indicates stronger discriminative capability, with values closer to one representing better classification performance.

Accuracy was calculated as the proportion of correctly classified customers among all observations. Although accuracy is easy to interpret, it may produce overly optimistic results when one class dominates the dataset.

Precision measures the proportion of customers predicted to repurchase who actually returned. High precision reduces unnecessary marketing expenditure because fewer promotional resources are allocated to customers who are unlikely to make another purchase.

Recall represents the proportion of actual returning customers that were successfully identified by the model. From a business perspective, recall is particularly important because failing to identify valuable returning customers may reduce the effectiveness of customer retention campaigns.

Finally, the F1-score combines Precision and Recall into a single metric by calculating their harmonic mean. Since this project focuses on customer retention with moderately imbalanced data, the F1-score provides a balanced measure of prediction performance. Therefore, together with ROC-AUC, the F1-score was regarded as the primary evaluation criterion throughout this study.

8. Results and Evaluation

8.1 Model Performance Comparison

Both machine learning models achieved satisfactory predictive performance on the validation dataset.

Overall, Logistic Regression slightly outperformed Random Forest across all three evaluation metrics. Although the differences were relatively small, Logistic Regression consistently achieved higher ROC-AUC and F1-score values, indicating a better balance between correctly identifying customers who would repurchase and reducing false positive predictions. Furthermore, Logistic Regression offers greater model interpretability, making it more appropriate for practical business applications where transparent decision-making is essential.

8.2 ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curves provide a visual comparison of the classification performance achieved by the two models across different decision thresholds. As shown in Figure X, the ROC curve of Logistic Regression remains consistently above that of Random Forest, confirming its slightly stronger discriminative capability.

The obtained ROC-AUC value of approximately 0.75 indicates that the model can effectively distinguish customers who are likely to repurchase from those who are unlikely to return. Although Random Forest is capable of modelling nonlinear relationships, its additional complexity did not produce superior predictive performance on the current dataset. This finding suggests that customer repurchase behaviour can largely be explained by the engineered behavioural features using a relatively simple linear classification model.

8.3 Customer Value Pyramid Analysis

To further interpret the prediction results from a business perspective, customers were segmented into four value tiers according to their historical Monetary values. For each customer segment, the predicted repurchase probability was compared with the observed repurchase rate.

The analysis reveals a clear upward trend across customer value segments. Customers belonging to higher-value tiers exhibit both higher actual repurchase rates and higher predicted probabilities than lower-value customers. This finding supports a fundamental assumption of customer relationship management, namely that customers who have historically spent more are generally more likely to make future purchases.

Moreover, the close agreement between predicted probabilities and observed repurchase rates indicates that the developed prediction model successfully captures meaningful behavioural patterns. From a business perspective, customers in the highest-value segments should receive priority in marketing campaigns because they offer both greater expected revenue and stronger retention potential.

8.4 Feature Importance Analysis

Feature importance was analysed using the Random Forest model to better understand which variables contributed most to prediction performance. The results indicate that Monetary, Frequency, Monetary per Frequency, Recency, Average Order Value, Maximum Order Value, Number of Purchased Products, and Product Cluster were among the most influential predictors.

Among these variables, Monetary and Frequency contributed the greatest predictive power, suggesting that customers with higher cumulative spending and more frequent purchasing behaviour are significantly more likely to make another purchase. Recency also played an important role, with customers who purchased more recently exhibiting substantially higher repurchase probabilities than those whose last purchase occurred further in the past.

Interestingly, the product preference clusters generated through TF-IDF and K-Means also appeared among the most important variables. Although their contribution was smaller than that of the traditional RFM features, they provided additional behavioural information that enhanced the overall predictive capability of the model.

8.5 Business Interpretation

Beyond predictive accuracy, the developed models provide several valuable business insights. First, historical purchasing behaviour remains the strongest indicator of future purchasing decisions, confirming the effectiveness of behavioural features such as RFM variables in customer retention prediction. Second, customer segmentation based on behavioural characteristics enables companies to allocate marketing resources more efficiently by targeting customers with the highest expected return.

Finally, the experimental results demonstrate that relatively simple machine learning models can achieve competitive predictive performance when supported by carefully engineered customer features. This finding suggests that organisations do not necessarily require highly complex algorithms to obtain useful business predictions, provided that meaningful behavioural information is extracted from historical transaction data.

9. Model Deployment

Developing an accurate prediction model represents only one component of a successful data science project. To generate practical business value, the model must be integrated into operational business processes where prediction results can support marketing decisions in real time. Therefore, a deployment framework was designed to demonstrate how the developed customer repurchase prediction model could be incorporated into an online retail marketing system.

9.1 Deployment Architecture

The proposed deployment pipeline consists of six sequential stages, beginning with transaction data collection and ending with marketing decision support. New transaction records are continuously collected from the retailer's sales platform. After data cleaning and feature extraction, customer-level behavioural features, including RFM variables and additional statistical indicators, are automatically updated whenever new transactions are received.

The processed customer features are subsequently stored in a customer feature database and passed to the trained Logistic Regression model. The model predicts the probability that each customer will make another purchase within the next thirty days. Based on these prediction scores, customers are classified into different marketing segments that support personalized promotional strategies.

9.2 Business Application

The prediction results can directly support customer relationship management and personalized marketing activities. Customers with high predicted repurchase probabilities may receive personalized discount coupons, product recommendation emails, loyalty rewards, or early access to promotional campaigns. These incentives encourage repeat purchases while strengthening long-term customer relationships.

Customers with medium repurchase probabilities may instead receive reminder emails or seasonal promotional messages designed to stimulate additional purchases. Conversely, customers with low predicted probabilities may be excluded from expensive marketing campaigns, enabling companies to allocate promotional budgets more efficiently. Such targeted marketing strategies reduce unnecessary advertising expenditure while improving overall marketing return on investment.

9.3 System Maintenance

Customer purchasing behaviour changes continuously over time, making regular model maintenance essential for maintaining prediction accuracy. To ensure stable long-term performance, transaction data should be updated on a daily basis, while customer-level behavioural features should be recalculated automatically whenever new data become available.

In addition, model performance should be continuously monitored using evaluation metrics such as ROC-AUC and F1-score. The prediction model is recommended to be retrained monthly or quarterly using newly collected transaction records. Comparing model performance before deployment and after retraining enables organisations to detect performance degradation and adapt the prediction system to evolving customer behaviour.

10. Limitations

Although the developed models achieved satisfactory predictive performance, several limitations should be acknowledged.

10.1 Limited Temporal Validation The available dataset contains only one year of transaction history. Consequently, chronological train-test splitting was not feasible, and a random train-validation split was adopted instead. Random splitting may introduce information leakage because purchase behaviour observed in later periods can influence earlier predictions. As a result, the reported performance may be slightly more optimistic than would be expected in a real-world deployment environment. Future studies should employ time-series validation techniques to better simulate practical forecasting scenarios.

10.2 Limited Data Coverage The dataset used in this project represents transaction records from a single online retailer over approximately one year. Customer purchasing behaviour may vary substantially across industries, geographical regions, and economic environments. A longer observation period would enable seasonal purchasing patterns and long-term behavioural trends to be modelled more accurately. Furthermore, incorporating additional external variables, such as promotional activities, holidays, and macroeconomic indicators, may further improve prediction performance.

10.3 Class Imbalance Similar to many customer retention problems, customers who did not make another purchase constituted the majority class in the dataset. Although balanced class weights were applied during model training to reduce prediction bias, class imbalance may still influence classification performance.

Future research could investigate more advanced techniques for handling imbalanced datasets, including Synthetic Minority Over-sampling Technique (SMOTE), cost-sensitive learning, classification threshold optimisation, and ensemble sampling methods. These approaches may further improve the model's ability to identify customers with high repurchase potential while maintaining overall prediction accuracy.

11. Future Work

Several directions could further improve this project.

First, more advanced machine learning algorithms such as XGBoost, LightGBM, or CatBoost could be evaluated to determine whether nonlinear boosting methods provide superior predictive performance.

Second, richer behavioural features could be developed. Examples include customer purchase intervals, seasonal purchasing preferences, product diversity, and customer lifetime value (CLV).

Third, incorporating external information such as holiday calendars, promotional campaigns, customer demographics, or website browsing behaviour may significantly improve prediction accuracy.

Finally, the deployment framework could be extended into a real-time recommendation system capable of updating customer scores immediately after each transaction. Such a system would enable dynamic personalized marketing and more responsive customer engagement.

12. Conclusion

This project successfully completed an end-to-end data science workflow for customer repurchase prediction using real-world online retail transaction data.

The project began with comprehensive exploratory data analysis to understand customer purchasing behaviour and business patterns. Based on these insights, customer-level behavioural features were engineered using the RFM framework, statistical purchasing indicators, and product preference clusters.

Two supervised machine learning models, Logistic Regression and Random Forest, were developed and evaluated. Experimental results demonstrated that Logistic Regression achieved the best overall performance, with an ROC-AUC of approximately 0.75 while maintaining strong interpretability.

Feature importance analysis confirmed that historical spending, purchase frequency, and purchase recency are the strongest indicators of future customer repurchase behaviour.

To demonstrate practical applicability, a deployment framework was proposed that integrates the trained prediction model into an online retail marketing system. By automatically identifying customers with high repurchase probability, the proposed system supports targeted promotions, improves customer retention strategies, and enhances marketing efficiency.

Overall, this project illustrates how data science techniques can transform raw transaction data into actionable business intelligence. The combination of exploratory analysis, predictive modelling, and deployment planning provides a complete example of an end-to-end machine learning solution for customer relationship management.